

PORTFOLIO

DATA ANALYST

&

FOOTBALL PERFORMANCE

THOMAS GUIBERT

Data Analyst – Football, Performance & Career Development

Audencia & Centrale Nantes – Data Science & AI

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PROFESSIONAL PROFILE

Data analyst specializing in football performance, injury prevention and player valuation.

Trained in Data Science, Statistics and AI (Audencia x Centrale Nantes), I use data to support:

- **Football clubs**
- **Sporting directors**
- **Performance & medical staff**
- **Match analysts**
- **Player agents (en dernier)**

My expertise: **quantifying player progression, securing transfers, assessing return-to-play readiness, identifying performance factors, and building data-driven player profiles.**

I also work on performance monitoring, match-related data analysis, workload indicators, injury-risk modelling and functional profiling — areas directly relevant to performance departments and match analysis units within professional clubs.

Value for a club — Performance & Match Analysis

- Monitoring of physical and functional performance
- Identification of performance trends over time
- Data-driven support for return-to-play decisions
- Analysis of workload, asymmetries and risk indicators
- Visualisation dashboards for staff meetings
- Benchmarking of players within the squad
- Support for match analysis (event data, simple tracking insights)
- Clear communication of insights to coaches and performance staff

Value for a player agent

- Objectively track a **player's progression** after injury
- Provide **data-driven arguments** for negotiations
- Secure **transfers** with reliable, evidence-based insights

- Identify **functional strengths and weaknesses**
- Benchmark a player against **performance standards**
- Build a compelling, **data-enhanced player profile**
- Monitor a player's evolution over several months

SHORT ANALYSIS — DATA & FOOTBALL NEWS

When Alisson Becker Scores from a Corner: A Historic Premier League Moment

This analysis follows the approach recommended by a StatsPerform Data Scientist: producing short, reactive, football-related content to showcase analytical skills.

On May 16, 2021, Alisson Becker (Liverpool) scored a header in the 95th minute against West Bromwich Albion, from a corner kick.

This goal became historic because:

- First headed goal ever scored by a goalkeeper in Premier League history
- First Liverpool goalkeeper to score in an official match
- Scored in the 95th minute, with Champions League qualification at stake
- Occurred on an offensive corner with the goalkeeper joining the attack

Why is this analytically interesting?

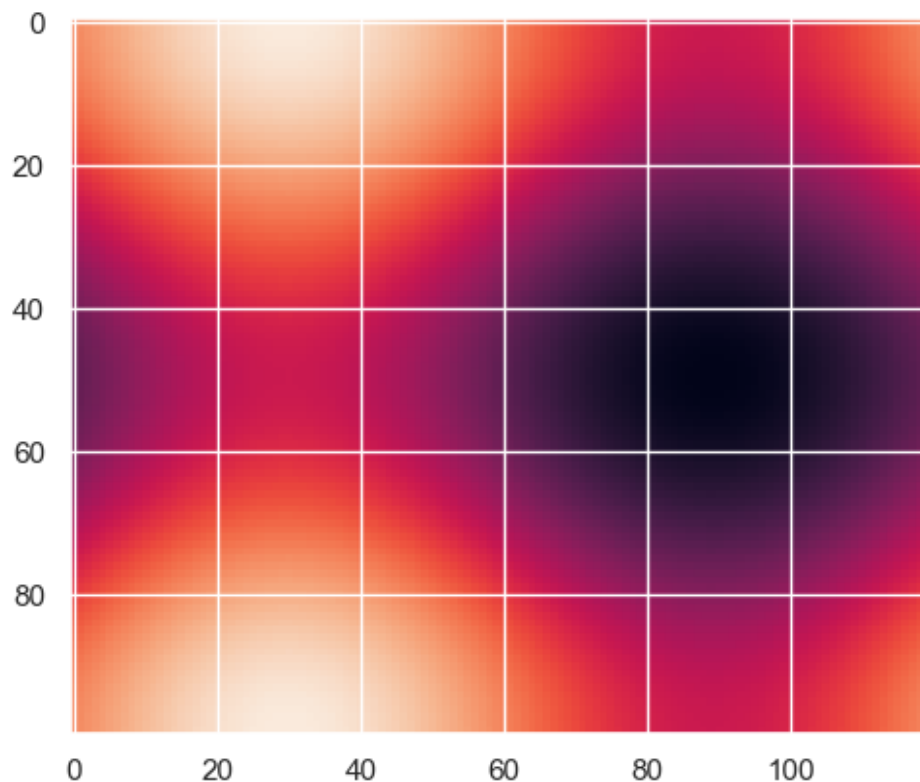
- Goals scored by goalkeepers are extremely rare
- The match context explains the high-risk decision
- The analysis can explore:
 - Historical frequency of goalkeeper goals
 - Tactical context of late-game goalkeeper involvement
 - Profiles of goalkeepers capable of scoring
 - Psychological impact and momentum shifts

Analytical value: This type of event connects data, tactical context and storytelling — a highly appreciated format in football analytics.

ADVANCED PROJECTS — ANALYSIS & VISUALIZATION

These projects were completed in 2024–2025, after my internship at Kinesport. They reflect a more advanced level in data science, visualization and football analytics.

Heatmap — Spatial Analysis (Python)



Objective: Analyze the spatial distribution of an indicator (intensity, danger, activity, etc.) on a modeled pitch grid.

Skills used:

- 2D matrices, interpolation
- Advanced visualization (Matplotlib / Seaborn)
- Data normalization
- Spatial analysis applied to football

Value for clubs / analysts: Helps visualize zones of influence, activity or danger for a player or a team. This type of spatial modelling can be applied to event data or tracking data to analyse zones of influence, pressing patterns or positional tendencies.

Python Animation — Spatiotemporal Movement Generation

```
Entrée [82]: 1 fig, ax = plt.subplots()
2
3 ims = []
4 for i in range(60):
5     x += np.pi / 15
6     y += np.pi / 30
7     im = ax.imshow(f(x, y), animated=True)
8     if i == 0:
9         ax.imshow(f(x, y))
10    ims.append([im])
11
12 ani = animation.ArtistAnimation(fig, ims, interval=50, blit=True, repeat_delay=1000)
13 ani.save("movie.gif")
```

Objective: Create a dynamic animation illustrating the evolution of a phenomenon (movement, intensity, trajectory) over time.

Skills used:

- Python (Matplotlib Animation)
- Frame-by-frame image generation
- Manipulation of evolving matrices
- Construction of analytical GIFs

Value for clubs / analysts: Useful for representing tracking-data sequences, collective movements or tactical evolutions. This approach can be used to illustrate player movements, collective behaviours or tactical sequences based on tracking data.

MATCH ANALYSIS — ANALYTICAL APPROACH

Objective: Explore match-related data to identify patterns, tendencies and key actions relevant to coaches and analysts.

Skills used:

- Identification of key actions and patterns
- Simple event-data exploration (passes, shots, recoveries)
- Visualisation of positional tendencies
- Linking physical and tactical indicators
- Clear reporting for coaches

Familiar with the analytical logic behind event data (passes, shots, recoveries, sequences) and with the structure of tracking-based spatial and temporal analyses used in professional clubs.

Value for clubs: Provides actionable insights for match analysis units, supports tactical discussions, and helps connect physical performance with on-field behaviours.

TECHNICAL SKILLS

Languages & Tools

Python (Pandas, NumPy, Matplotlib, Seaborn, Statsmodels, Scipy) Jupyter Notebook – SQL – Power BI – Advanced Excel

Statistics & Data Science

Statistical tests (T-test, Chi², ANOVA, Pearson) Regression models – Post-hoc analyses – EDA – Advanced visualization Machine learning preparation (neural networks)

Data Pipeline

Cleaning & merging – Missing-value handling Feature construction – Descriptive analysis – Modeling

Football & Field Expertise

10 years of competitive football — **Regional 1** level Occitanie Champion Strong knowledge of functional tests: Single Hop, Triple Hop, Side Hop, Crossover Hop, Illinois, ACL-RSI... Personal experience with ACL rupture (2023) On-field performance analysis

EDUCATION — KEY COURSES & GRADES

Technical Courses (Audencia x Centrale Nantes)

Scientific Python (Informatics I → VI) — 16 to 18/20

Mathematics (Analysis, Algebra, Probability & Statistics) — 17 to 18/20

Data Analysis & Business Intelligence — 12.7/20

Data Mining — 16/20

Machine & Deep Learning — 15.3/20

Natural Language Processing — 13.8/20

Scientific Visualization — 15.6/20

Cloud Computing — 16/20

Semester in Shenzhen (China)

Blockchain & Business Analytics — 19/20

Introduction to Fintech — 18.4/20

Chinese Language — 20/20

Internet of Things — 15.8/20

CASE STUDY — BIG DATA ANALYSIS OF THE KSTARTS TEST

Kinesport / Medinetic Learning — 11Stadium (Lyon) Role: Data Scientist Intern

Case study completed at the end of my first year. My skills have since evolved through advanced coursework in years 2 and 3.

Objective

Analyze **2,088 tests**, **266 variables** and **550,000+ data points** to:

- Assess a player's ability to return to previous performance levels
- Identify factors influencing functional performance
- Support **return-to-play**, **recruitment** or **transfer** decisions
- Provide data-driven arguments to **enhance player valuation**

KEY RESULTS

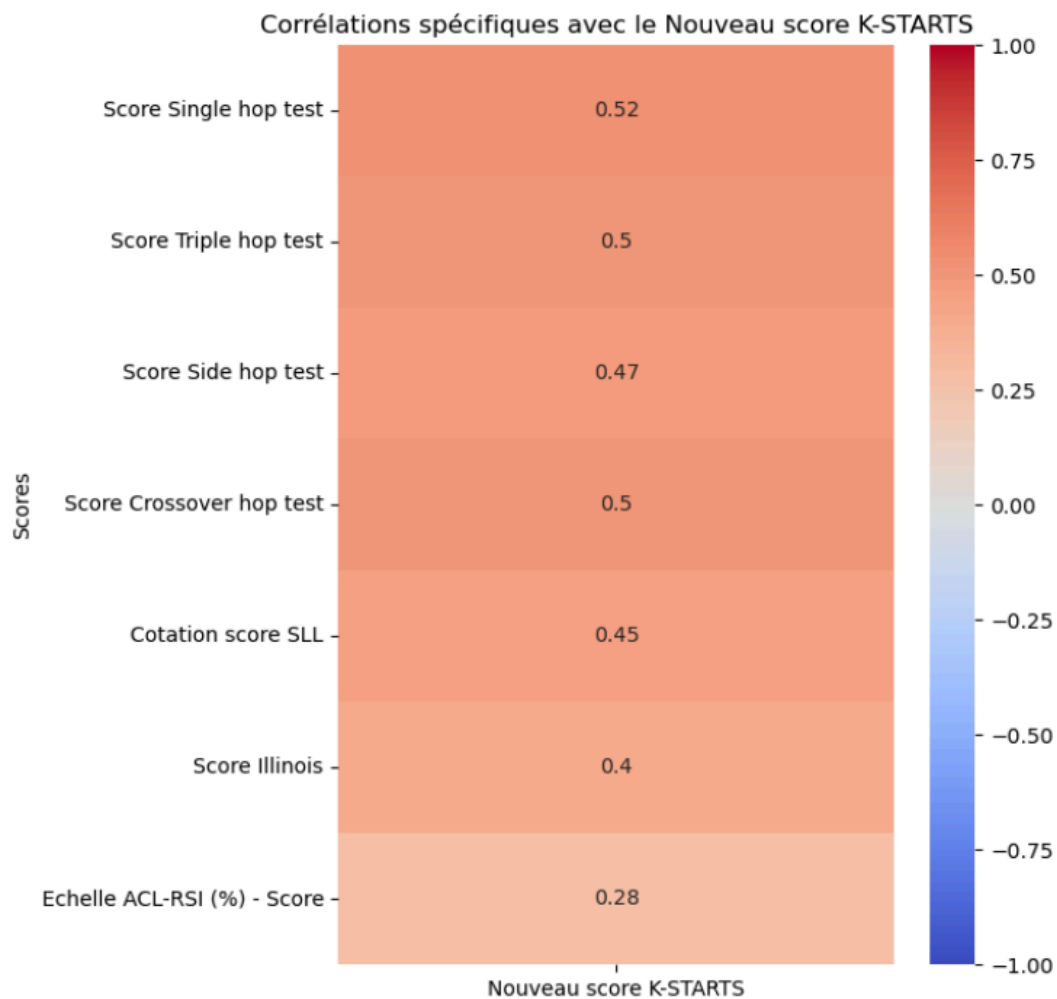
1. Side Hop is the strongest predictor of post-ACL performance

Heatmap of KSTARTS correlations

```
# Affichage de la matrice de corrélation spécifique avec un heatmap
plt.figure(figsize=(6, 8))
sns.heatmap(correlation_df, annot=True, cmap='coolwarm', vmin=-1, vmax=1)
plt.title("Corrélations spécifiques avec le Nouveau score K-STARTS")
plt.ylabel('Scores')
plt.show()
```

2. Psychological readiness (ACL-RSI) directly influences return-to-play

Correlations between functional tests and the KSTARTS score (ACL-RSI: 0.28)



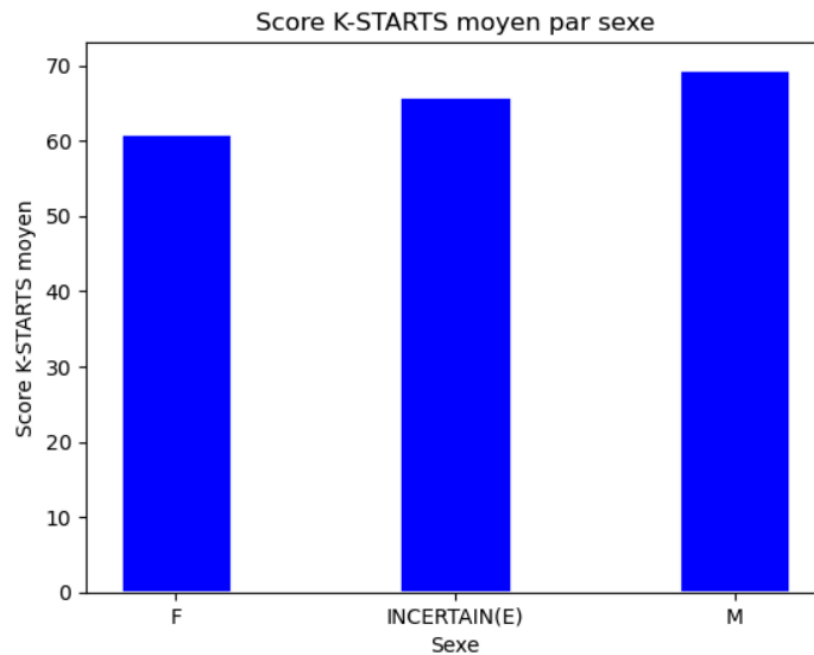
→ The psychological factor (ACL-RSI) shows a moderate correlation (0.28) with the KSTARTS score: psychologically ready players statistically tend to return faster and stronger. These insights support performance departments, medical teams and S&C coaches in return-to-play decisions and functional monitoring.

Heatmap (7 tests)

```
# Affichage de la matrice de corrélation spécifique avec un heatmap
plt.figure(figsize=(6, 8))
sns.heatmap(correlation_df, annot=True, cmap='coolwarm', vmin=-1, vmax=1)
plt.title("Corrélations spécifiques avec le Nouveau score K-STARTS")
plt.ylabel('Scores')
plt.show()
```

3. Male players achieve significantly higher scores

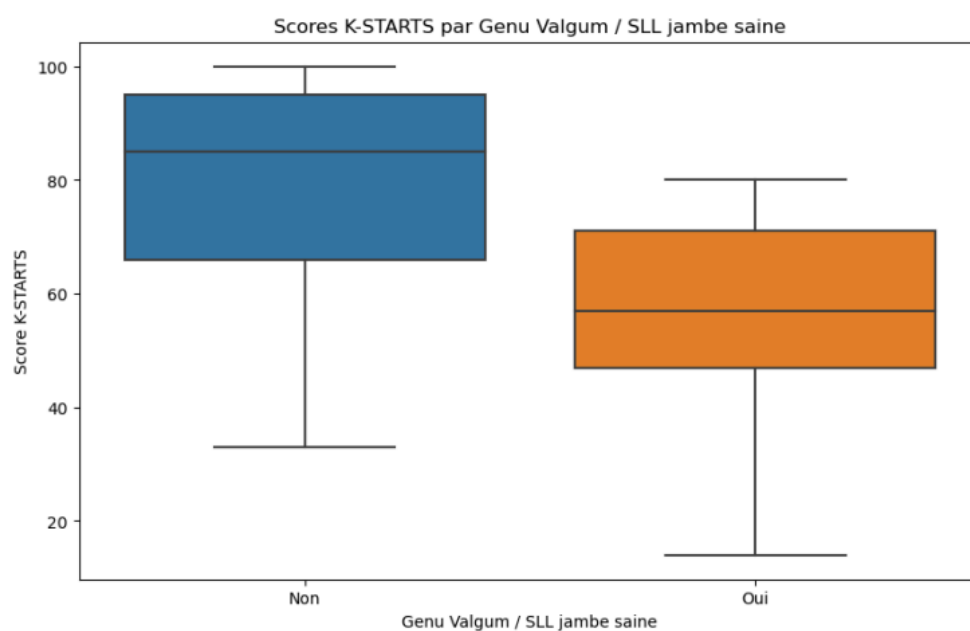
Boxplot — Sex vs KSTARTS Score



→ Helps adjust expectations based on player profile.

4. Genu valgum is a major negative factor

Boxplot — Genu Valgum vs KSTARTS Score



→ Players without valgum have a functional advantage — useful to reassure clubs about biomechanical stability.

5. Playing level influences functional performance

Pivot table — Playing Level vs KSTARTS Score

```
dfp=df.pivot_table(columns = 'Niveau du sport pratiqué', values='Score K-STARTS'  
display(dfp)
```

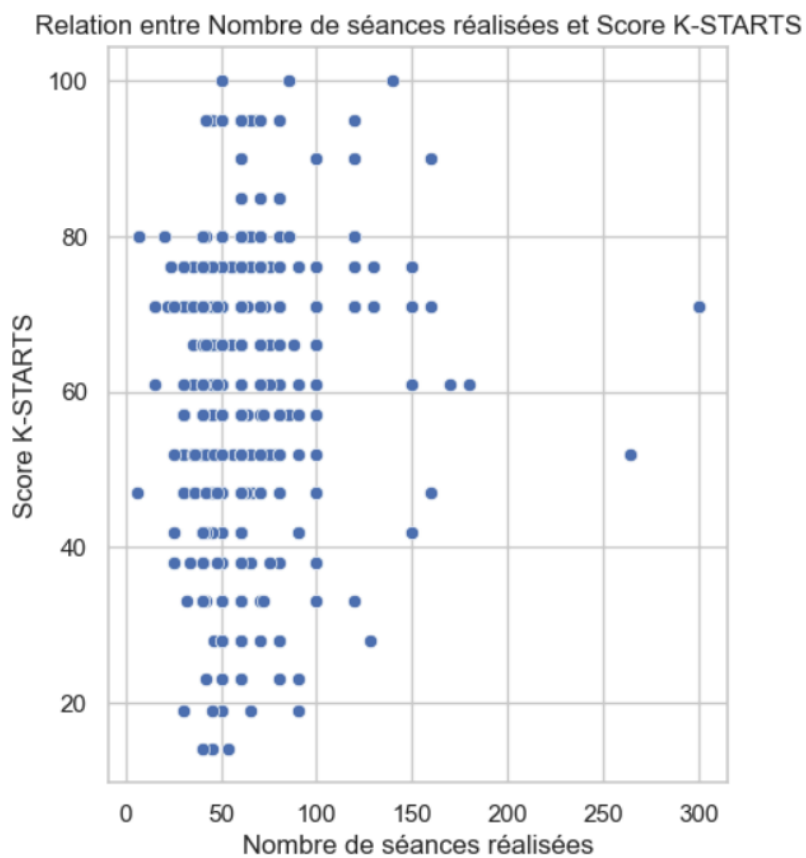
Niveau du sport pratiqué	Aucun	Départemental	International	Je ne sais pas	Loisir	National	Régional
Score K-STARTS	46.461538	56.414141	71.225806	67.363636	54.027397	67.395833	59.96



→ Higher-level players perform better — useful for benchmarking.

6. Number of rehabilitation sessions correlates with final score

Scatterplot — Sessions vs KSTARTS Score



→ More consistent rehabilitation leads to faster progression — a strong argument to demonstrate a player's commitment.

VALUE FOR PLAYER AGENTS

To enhance player valuation

- Highlight functional strengths
- Provide clear visuals for player dossiers
- Compare players to performance standards

To secure transfers

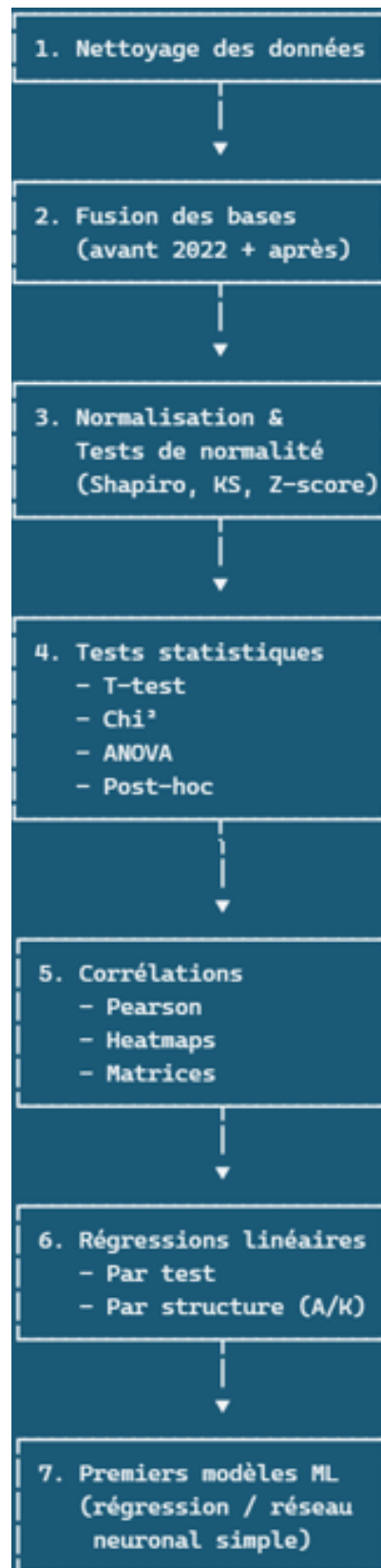
- Objective analysis of return-to-play
- Identification of residual risks
- Reliable data to reassure clubs

To support career management

- Longitudinal performance monitoring
- Identification of progression areas
- Construction of a complete data-driven profile

METHODOLOGY — DATA PIPELINE

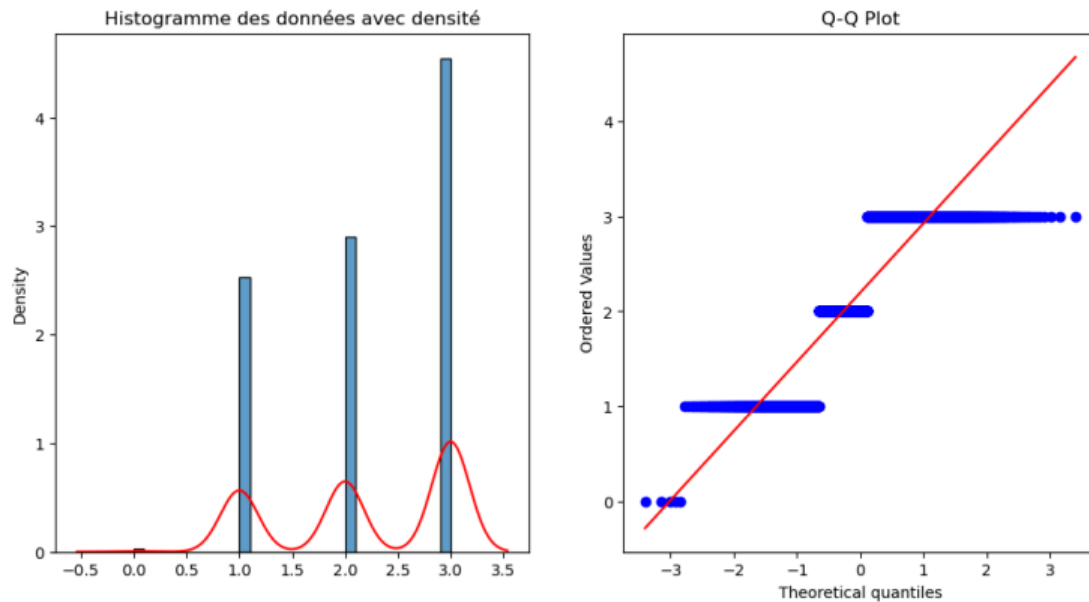
Pipeline diagram



STATISTICAL ANALYSES

Normality analysis

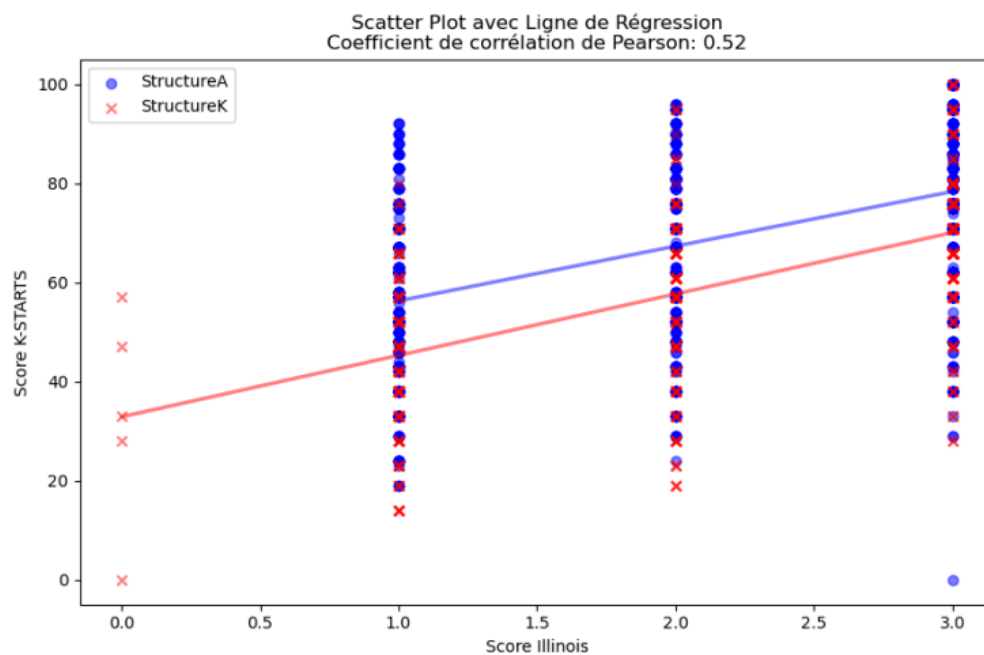
QQ Plot + Illinois Score Histogram



→ Justifies statistical choices.

Linear regressions

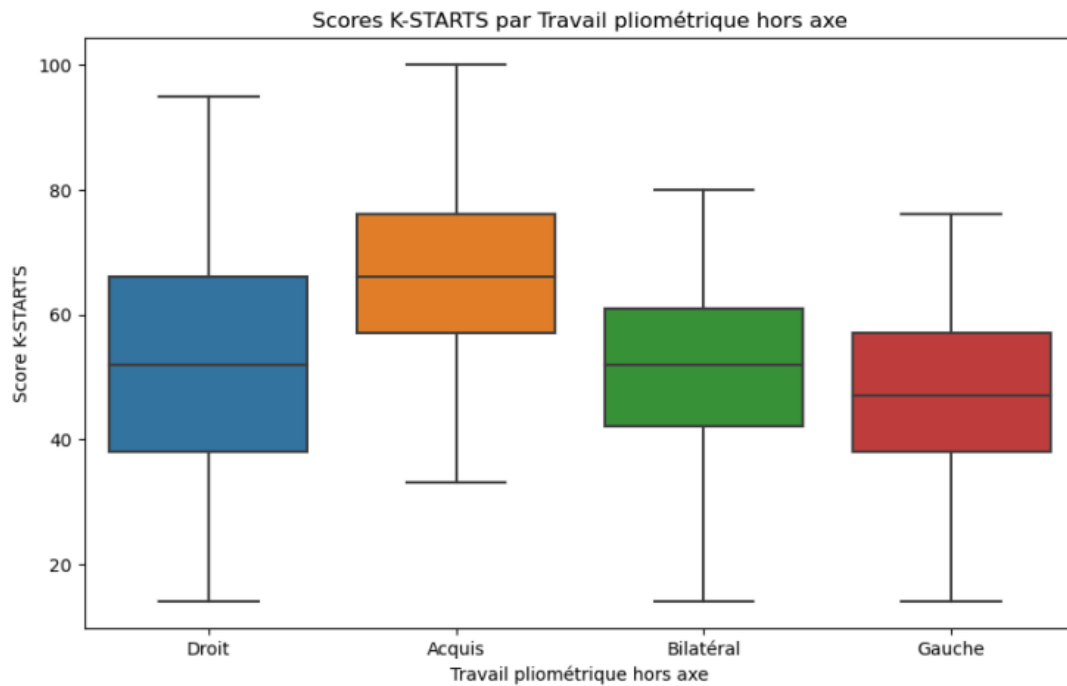
Regression — Illinois Score → KSTARTS Score



→ Agility is a strong predictor of performance.

ANOVA

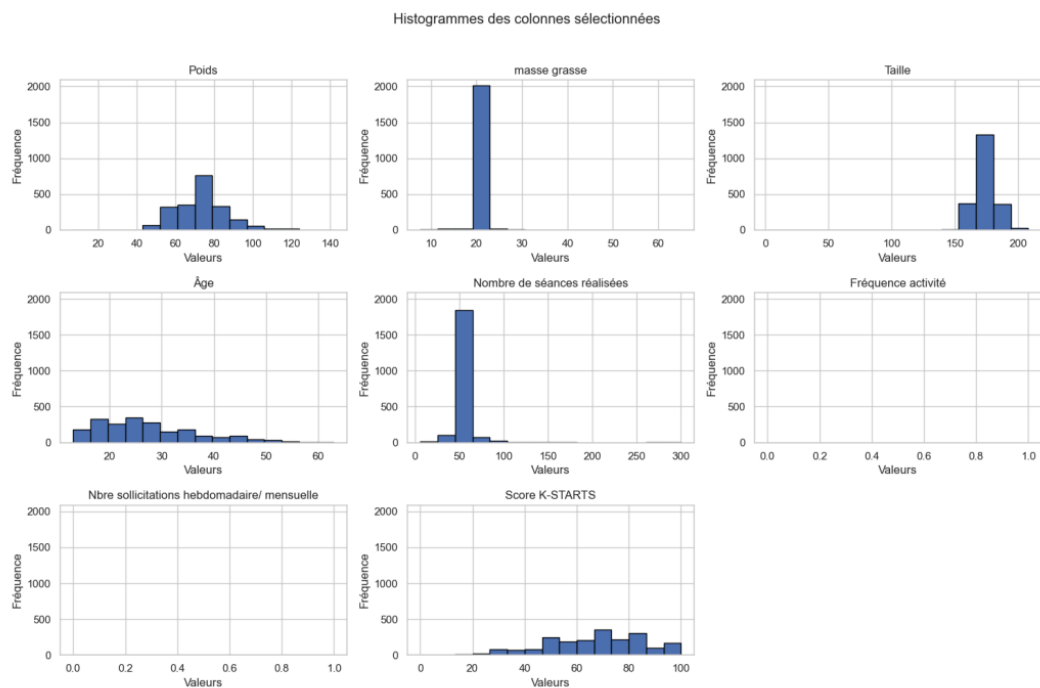
Boxplot — Off-axis plyometric work

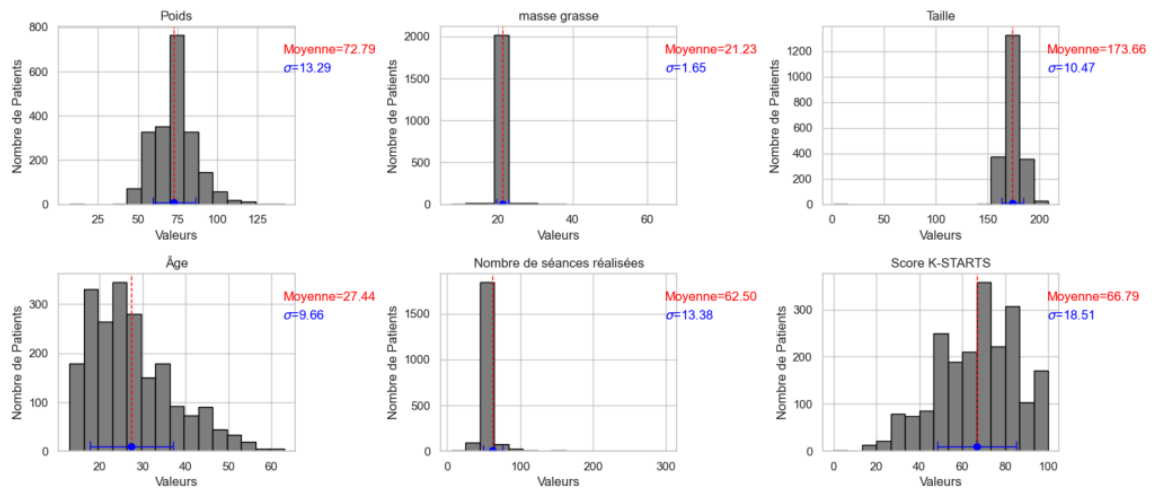


→ Shows the influence of training type on performance.

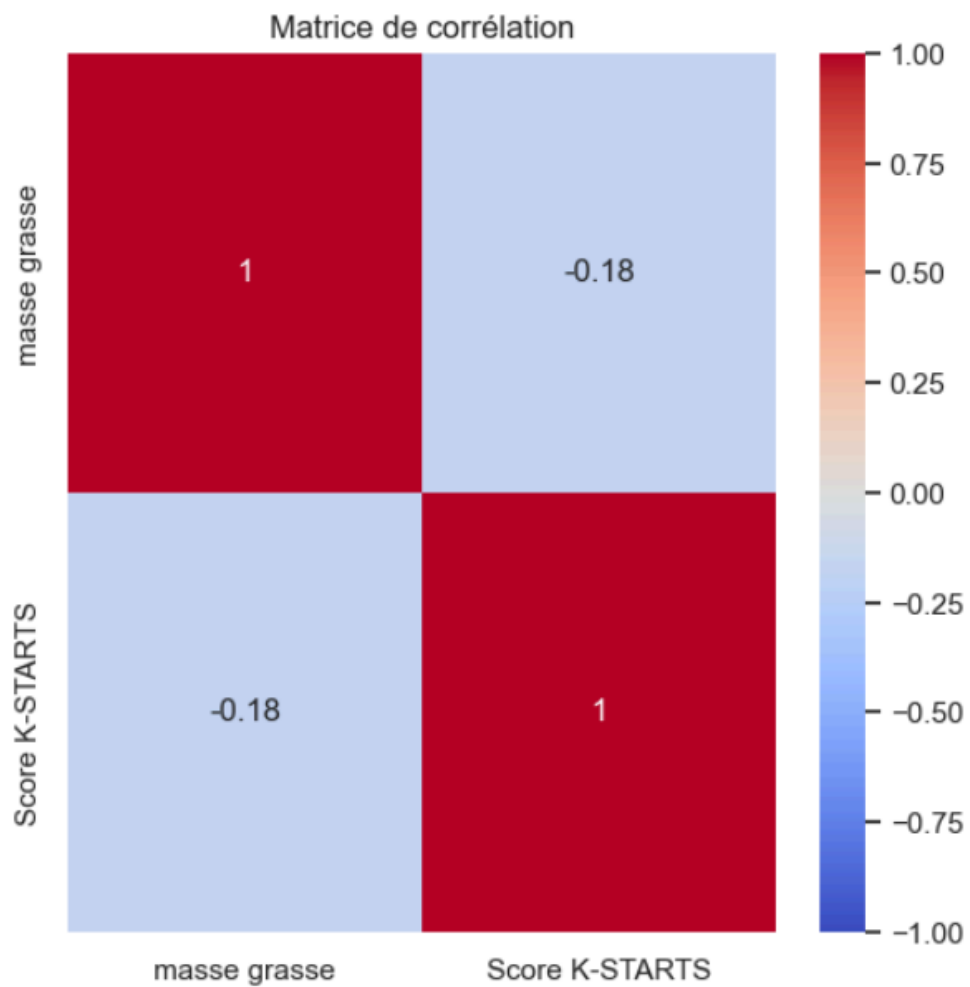
VISUAL ANNEX

- Histograms: Weight / Age / Height / Sessions

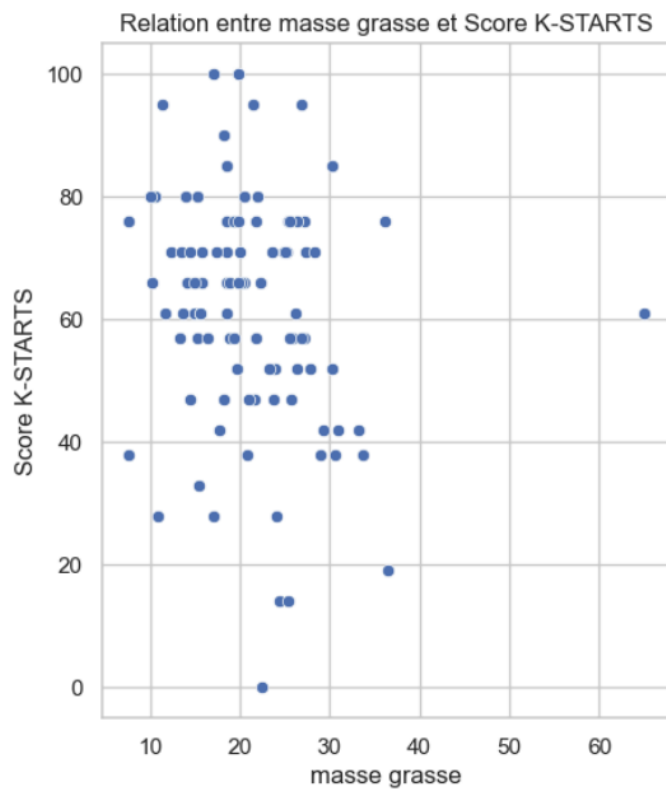
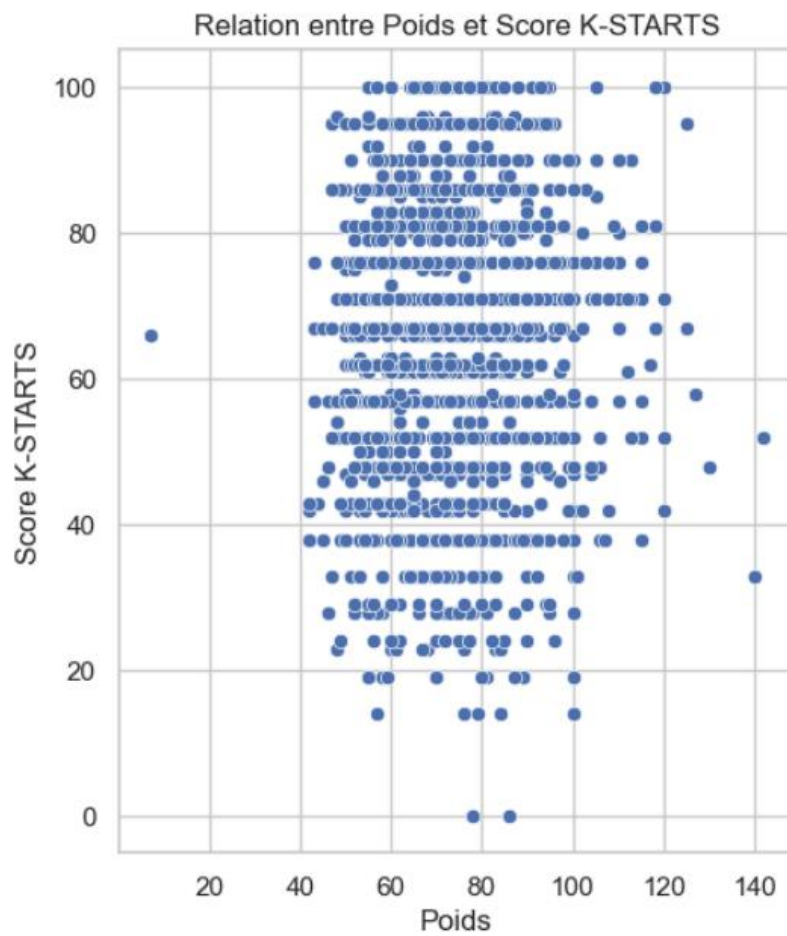


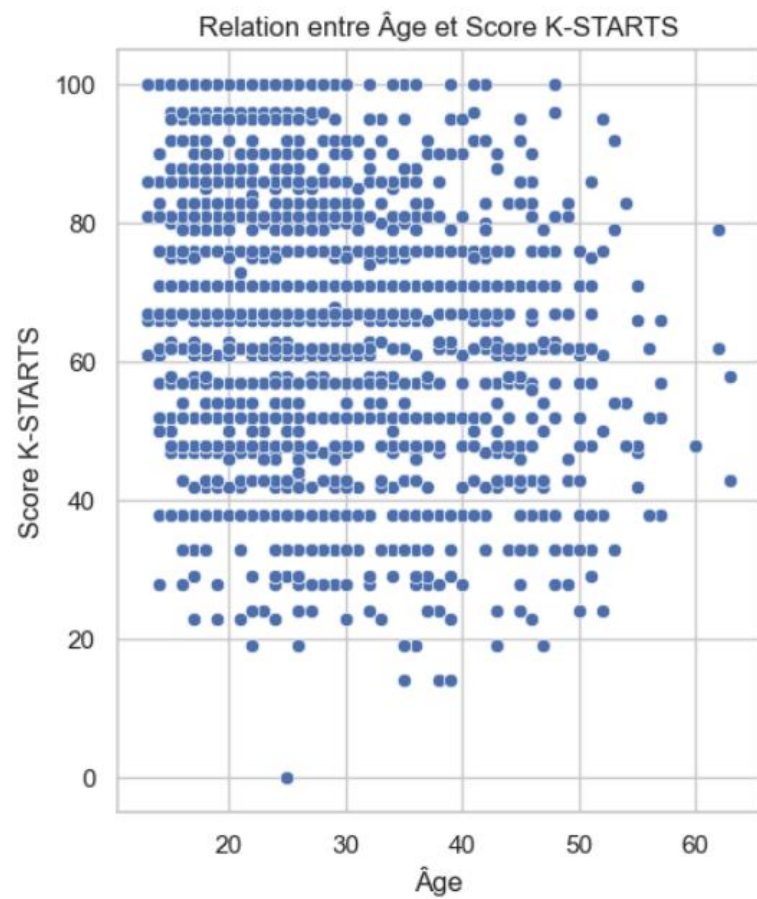
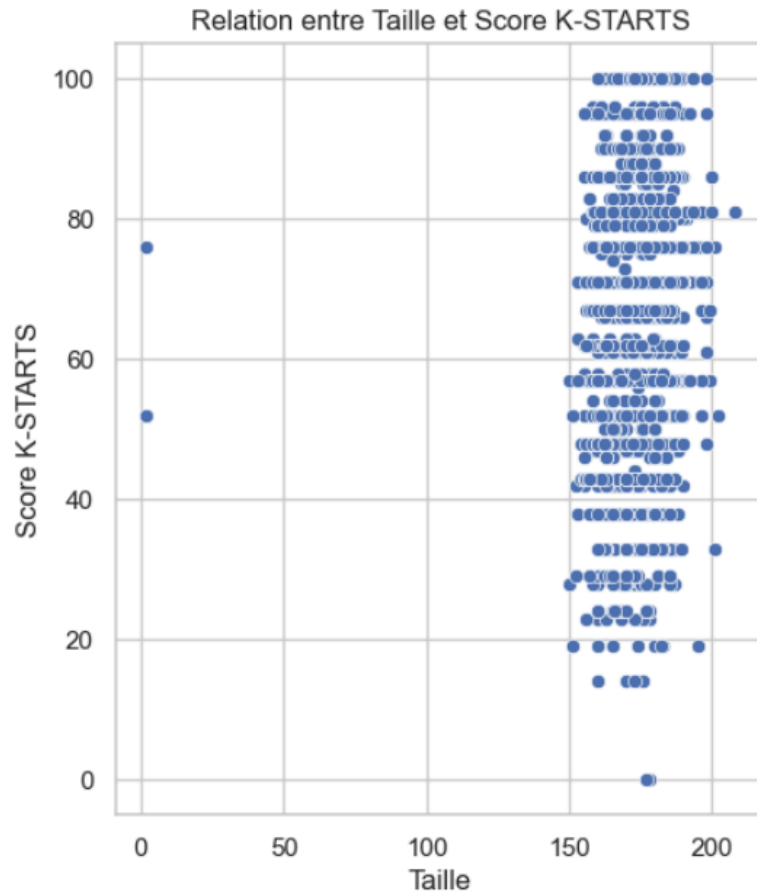


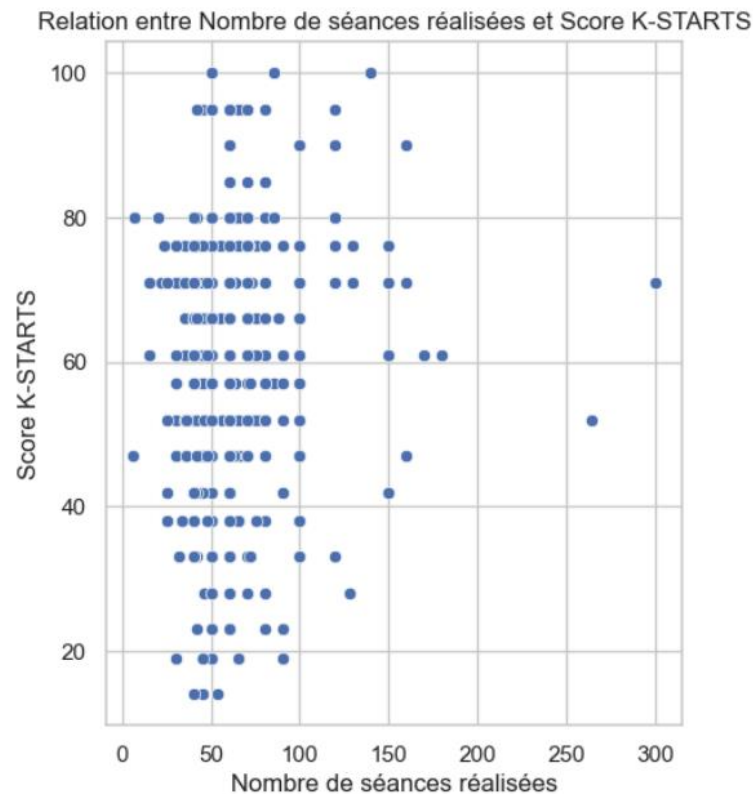
- Heatmap: Body fat vs KSTARTS Score



- Various scatterplots







PYTHON CODE EXTRACTS

- Statistical tests

```
# Test de Shapiro-Wilk
print("Test de Shapiro-Wilk")
shapiro_test = stats.shapiro(data)
print(f"Shapiro-Wilk test: statistic={shapiro_test.statistic}, p-value={shapiro_test.pvalue}")
if shapiro_test.pvalue > 0.05 :
    print("la p-value du test est >0.05 donc l'hypothèse nulle n'est pas rejetée")
    print("On conclue que la variable suit bien une loi normale : normalité vérifiée")
else:
    print("la p-value du test est < 0.05 donc l'hypothèse nulle est rejetée")
    print("On conclue que la variable NE suit PAS une loi normale : normalité non vérifiée")

print()

# Test de Kolmogorov-Smirnov
print("Test de Kolmogorov-Smirnov")
ks_test = stats.kstest(data, 'norm')
print(f"Kolmogorov-Smirnov test: statistic={ks_test.statistic}, p-value={ks_test.pvalue}")
if ks_test.pvalue > 0.05 :
    print("la p-value du test est >0.05 donc l'hypothèse nulle n'est pas rejetée")
    print("On conclue que la variable suit bien une loi normale : normalité vérifiée")
else:
    print("la p-value du test est < 0.05 donc l'hypothèse nulle est rejetée")
    print("On conclue que la variable NE suit PAS une loi normale : normalité non vérifiée")
```

- Correlations / régressions

```
# 2. Matrice de corrélation
```

```
plt.figure(figsize=(6,6))
correlation_matrix = df_filtered[columns_of_interest].corr(numeric_only=True)
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', vmin=-1, vmax=1)
plt.title("Matrice de corrélation")
plt.show()
```

```
plt.figure(figsize=(6,6))
columns_of_interest=["masse grasse", "Score K-STARTS"]
df_filtered=df[columns_of_interest]
correlation_matrix = df_filtered[columns_of_interest].corr(numeric_only=True)
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', vmin=-1, vmax=1)
plt.title("Matrice de corrélation")
plt.show()
```

```
# Calculer le coefficient de corrélation de Pearson
```

```
df_clean = df.replace([np.inf, -np.inf], np.nan).dropna(subset=['Taille', 'Score K-STARTS'])
corr_coefficient, p_value = pearsonr(df_clean['Taille'], df_clean['Score K-STARTS'])
print(f"Coefficient de corrélation de Pearson: {corr_coefficient}")
print(f"Valeur p: {p_value}")
```

```
# Calcul du coefficient de corrélation de Pearson en inversant les colonnes
```

```
df_clean = df.replace([np.inf, -np.inf], np.nan).dropna(subset=['Score Illinois', 'Score K-STARTS'])
corr_coefficient, p_value = pearsonr(df_clean['Score K-STARTS'], df_clean['Score Illinois'])
print(f"Coefficient de corrélation de Pearson: {corr_coefficient}")
print(f"Valeur p: {p_value}")
```

- Visualizations

```
# 2. Matrice de corrélation
```

```
plt.figure(figsize=(6,6))
correlation_matrix = df_filtered[columns_of_interest].corr(numeric_only=True)
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', vmin=-1, vmax=1)
plt.title("Matrice de corrélation")
plt.show()
```

```
# Créer Le graphique
```

```
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df_filtered, x='Taille', y='Score K-STARTS', hue='Sexe', style='Age')
```

```
# Histogramme avec densité
```

```
plt.subplot(1, 2, 1)
sns.histplot(data, bins=30, edgecolor='k', alpha=0.7, stat='density')
sns.kdeplot(data, color='red')
plt.title('Histogramme des données avec densité')
```

REFERENCES

“Thomas demonstrated professionalism, rigor and proactivity.” — **David Laplaud, PhD, R&D Manager, Medinetic Learning**

“A serious, pleasant student with strong analytical abilities.” — **Pierre Guiheneuc, Physiology Professor**

“Hard-working, attentive, always progressing.” — **Christophe Molina, Strength Coach**

CURRICULUM VITAE



✉ thomasguibert.etudes@gmail.com

🏠 Nantes & Toulouse, France

📅 20 years old

🌐 www.linkedin.com/in/thomas-guibert-big-data

🇫🇷 French

🚗 Driving licence (C)

☎ +33 7 68 09 17 39

Skills

Technical skills

Python (Pandas, NumPy, Matplotlib, Statsmodels), SQL, Jupyter Notebook - Excel, Power BI, Data visualisation - Statistics, Probability - RapidMiner

Languages

French : native language
English : professional proficiency
Spanish : intermediate
Mandarin : beginner

Interests

Football

10 years' practice, Regional level 1
Central defender & Left-back
Champion of the Occitanie Region & best U17 Régional 1 defence (2022)
Volunteer referee (June 2021)

Other sports

Rugby (2 years' practice) - Tennis (best 30.3 ranking) - Alpine skiing (gold medal, ESI) - Volleyball - Badminton - Squash - Bodybuilding - Pool - Darts

Video games

Sports, Fantasy Football, Racing, Battle Royale, FPS

Travel

North America, Europe, Asia, Africa

Additional certificates

Level 1 Civic First Aid & Prevention
WISC IV : 130

Thomas Guibert

3rd Year Student specialising in Data & AI — aiming to work in sport analytics and performance data

Seeking a 3 to 4-month internship in Data, AI & Management, available between 24 April and 14 August 2026, open to opportunities in France or internationally

Education

Bachelor of Business Administration Data, AI & Management

From September 2023 to June 2027 **AUDENCIA & CENTRALE NANTES** Nantes, France

English track / **Accreditations**: CTI, EUR-ACE, EQUIS, AACSB, AMBA • Centrale Nantes Top 250 • Audencia triple-accruited

- **Data / AI**: Python, Power BI, RapidMiner, scientific programming; data cleaning & quality, algorithms, data analysis; foundations in machine learning, deep learning & business-oriented AI
- **Management**: marketing, market research, innovation, sales techniques & negotiation; analytical, synthesis & communication skills
- **Other**: sports lessons, introduction to Mandarin

International Academic Semesters (as part of the BBA)

From February 2025 to January 2026

SHENZHEN UNIVERSITY & CENTRALE CASABLANCA China / Morocco

- **FinTech & emerging technologies**
Shenzhen University - China (Aug. 2025 → Jan. 2026)
FinTech, Internet of Things, blockchain, Chinese market & economy, business analytics
- **Data & AI**
Centrale Casablanca - Morocco (Feb. 2025 → May 2025)
AI & data mining for digital business, machine & deep learning, NLP, scientific visualization, information systems security

French Baccalaureate - with honours

From September 2020 to June 2023 **LYCEE JEAN-PIERRE VERNANT** Pins-Justaret, France

Mathematics, Physics & Chemistry

Sports option: rugby, athletics, badminton, bodybuilding

Oral exam: application of mathematics to Big Data in football (18/20)

Experience in Data & Sport

Junior Digital Strategy Analyst - Student project (ongoing)

From February 2026 to March 2026 **SAVEFOLIO** Nantes

- Working with a referral-tech startup to audit its digital presence and competitive landscape.
- Developing a full digital marketing strategy (audience, content, social media, SEO).
- Presenting strategic recommendations directly to the founder.

Data scientist intern

From June 2024 to August 2024 **KINESPORT / MEDINETIC LEARNING** Lyon

R&D at 11Stadium — European leader in sports physiotherapy

- Cleaned and merged athlete performance databases
- Conducted big-data exploration of K-STARTS functional test results
- Identified key factors influencing post-surgery recovery scores

Football Supervisor

November 2021 **BLOOMDAYS (formerly Côté Sports)** Toulouse, France

Led football sessions for 20 children (ages 6–10) and collected engagement data.

Developed communication, pedagogy and observation skills

Other experience

Retail assistant

July 2025 **SUPER U (SAS Eaudisse)** Eaunes

Admission officer

April 2024 **AUDENCIA** Nantes

Polyvalent Team Member

July 2023 **Mc DONALD'S** Portet-sur-Garonne, France